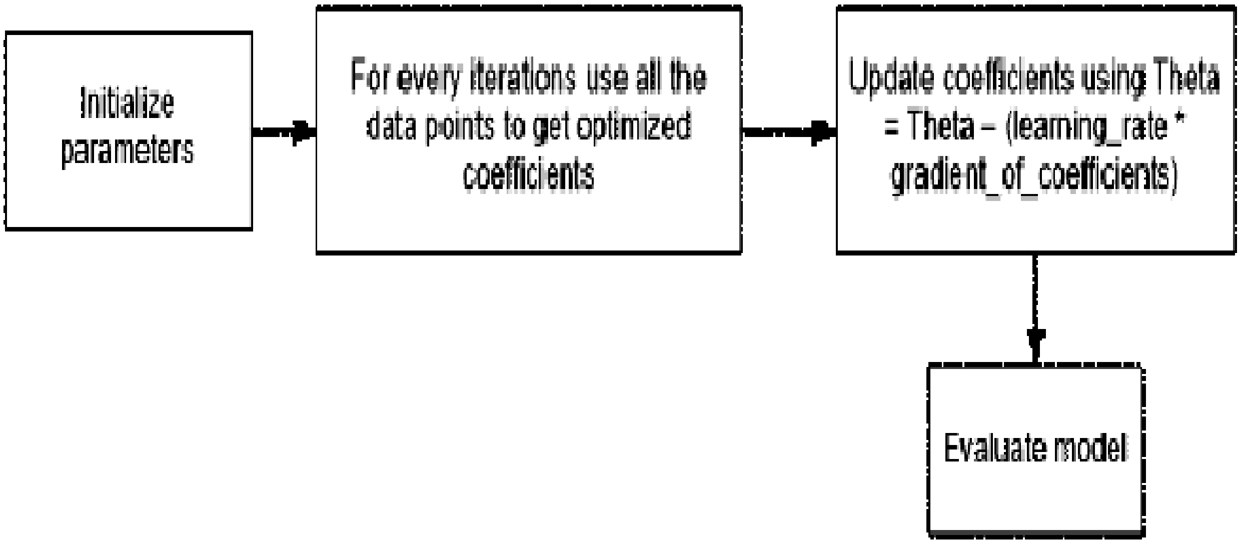
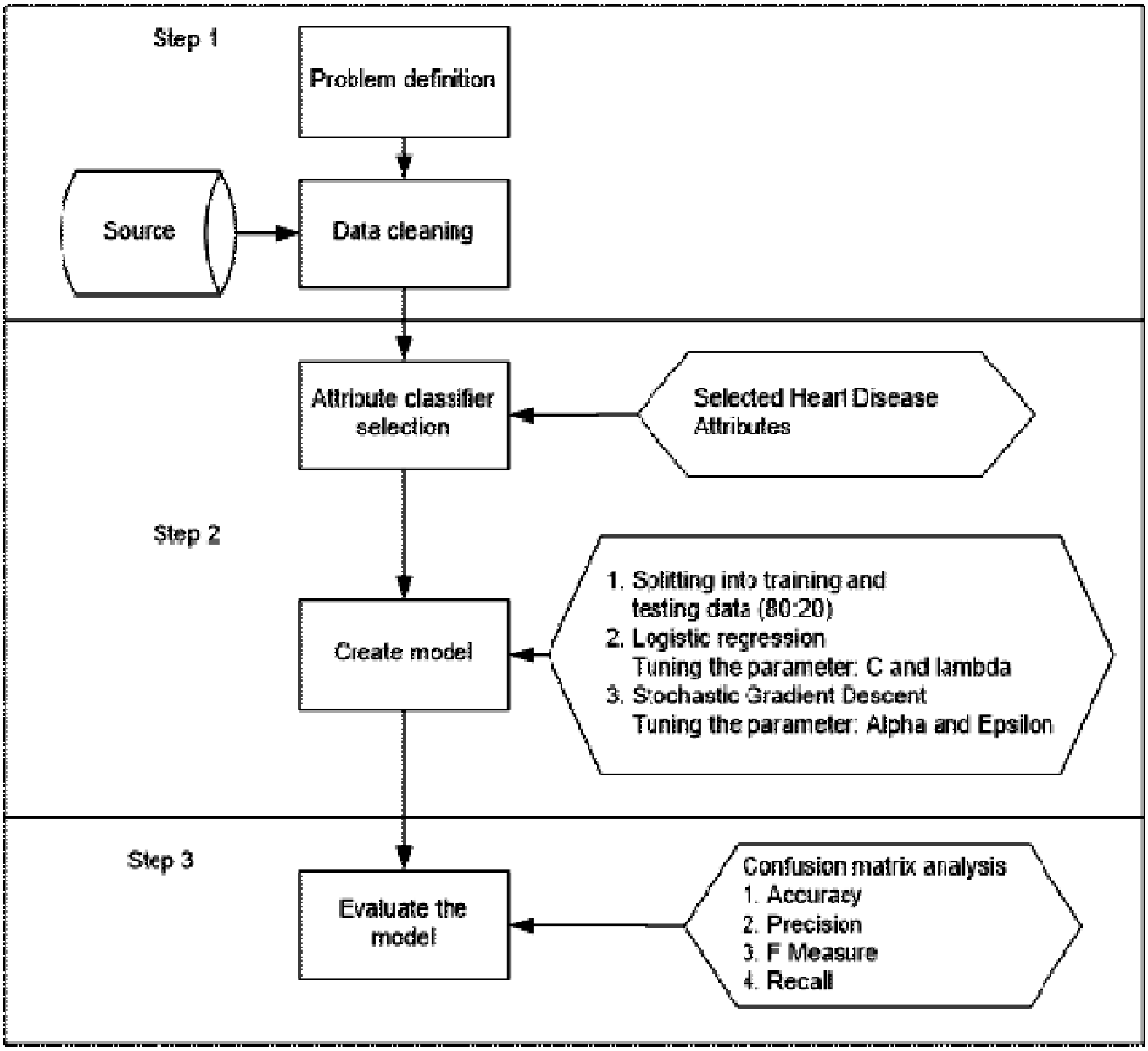

Initialize parameters
For every iterations use all the data points to get optimized coefficients
Update coefficients using Theta = Theta – (learning_rate * gradient_of_coefficients)
Evaluate model

SGD



Flowchart of Experiment

Q1. Select type and descriptions of dataset.

Q2. Take Logistic regression in the phyton library. Parameter tuning for optimizing classifier performance set for several parameter values namely: C =1.0, λ = 1.2 C .The default value is 1.0. The smaller C value specifies the stronger regularization value. The Lambda parameter used to control the regularization strength. The increase in the lambda values the strength of the regularization effect.

Q3. Find confusion matrix using logistic regression without SGD.

Q4. Find ACCURACY, PRECISION, F-MEASURE, RECALL: LOGISTIC REGRESSION CLASSIFIER

Q5. Learning rate that is derived from alpha and epsilon value. The learning rate managed the number of modification for estimating error. The alpha parameter is a constant value that multiplies the regularization term. The higher the value, the stronger the regularization. The default value is 0.0001. Epsilon parameter controlled the threshold value for getting precise prediction. Any distinction between the current prediction and the correct value is unobserved when this value less than the threshold value. The default value is 0.1. Experiment on this study set alpha value of 0.0001 and epsilon value of 0.1.

Q6. . Find confusion matrix using logistic regression with SGD.

Q7. . Find ACCURACY, PRECISION, F-MEASURE, RECALL: LOGISTIC REGRESSION CLASSIFIER with SGD.

Q8. Plot a graph Classifier result: Logistic regression and with Stochastic Gradient Descent for ACCURACY, PRECISION, F-MEASURE, RECALL.